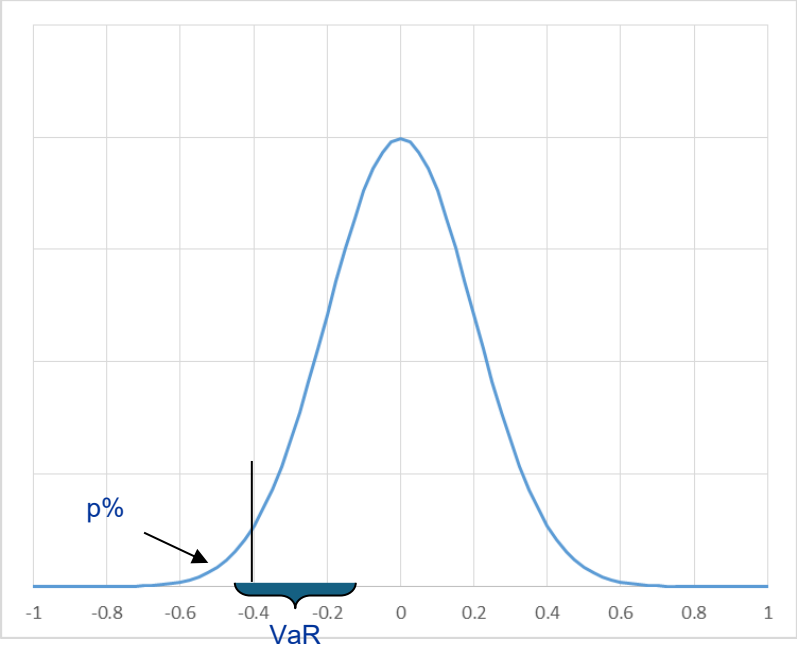
VALUE AT RISK

FINANCIAL ECONOMICS – SPRING 2026

THE NORMAL DISTRIBUTION

- How much can you lose at most over a certain period of time if you invest in a risky asset? Correct but useless answer: "100%". How can we arrive at an answer that can act as a guide for financial choices? Value at Risk (VaR) is the answer.

- The VaR allows a statement of the following type: "given an initial investment of value I, we are certain at p% that we will not lose more than €V in the next T days“. V is the Value at Risk (VaR): "the maximum loss that is exceeded, over the established time horizon, only with a small predetermined probability"



A €1 investment with a zero expected return, VaR is the maximum loss with pribability p%

HISTORY

- Introduced in the mid-1980s by JP Morgan analysts

- Even today we find the historical document (RiskMetrics Technical document) on the Internet

- Why the rapid spread of VaR?

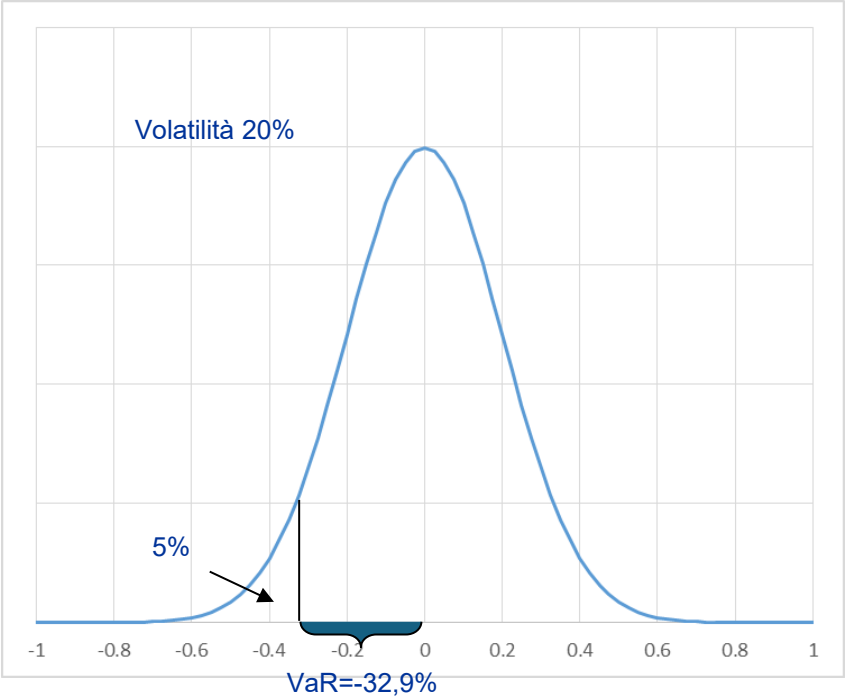
The possibility of conducting a stress test

The possibility of quantifying a loss in monetary terms

VAR EXAMPLE

- The assumption of normality of the return distribution allows us to calculate the VaR starting from the volatility of the return
Let's imagine an investment of 1 Euro (I=1) in a risky security with by zero average and 20% volatility over the 1-year time horizon

- Suppose we identify the "small probability" with 5%. From Excel we see that the return value that leaves a probability of 5% in the left tail of the distribution is equal to -32.9% NORM. INV(5%; 0; 20%)=32.9%. VaR is 32.9%. On an investment of €1, with an annual horizon, more than 32.9 cents are lost in only 5% of cases

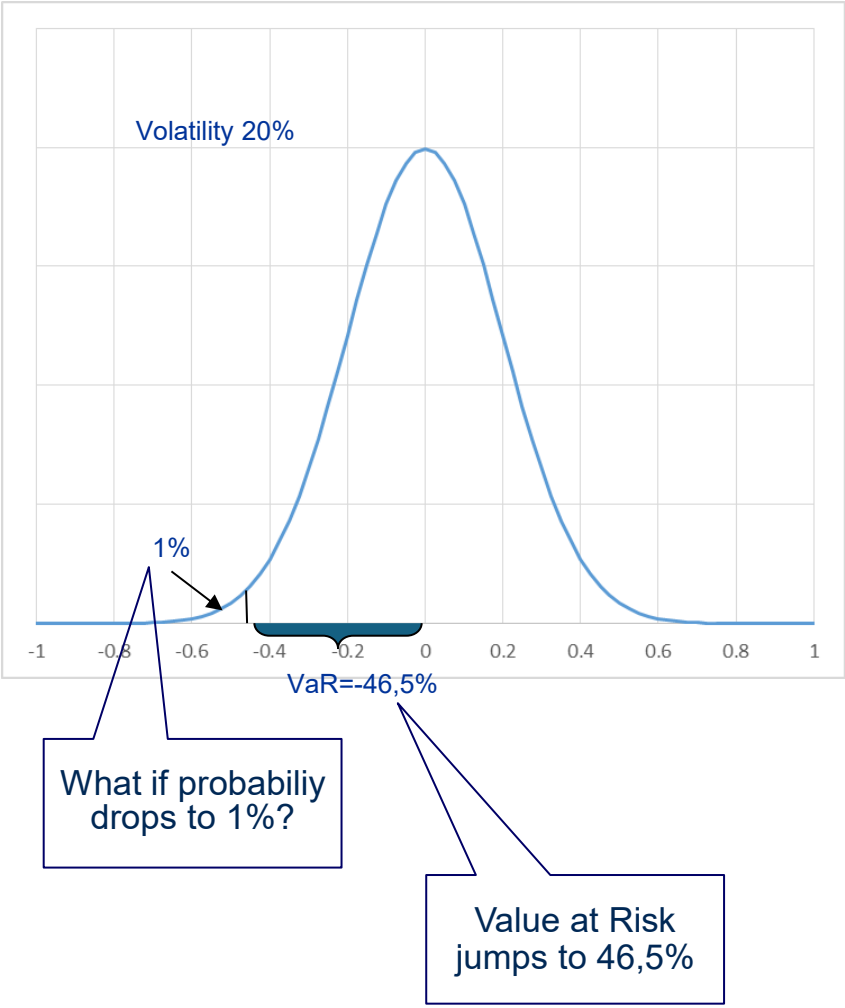


SENSITIVITY

- Let's imagine an investment of €1 (I=1) in a risky security with a probability distribution characterized by zero average and 20% volatility over the 1-year time horizon

- Suppose we identify the "small probability" with 1. From Excel we see that the return value that leaves a probability of 1% in the left tail of the distribution is equal to -46.5%. VaR is 46.5%

- On an investment of 1 Euro, with an annual horizon, more than 46.5 cents are lost in only 1% of cases. A more conservative probability of loss is associated with a greater potential loss

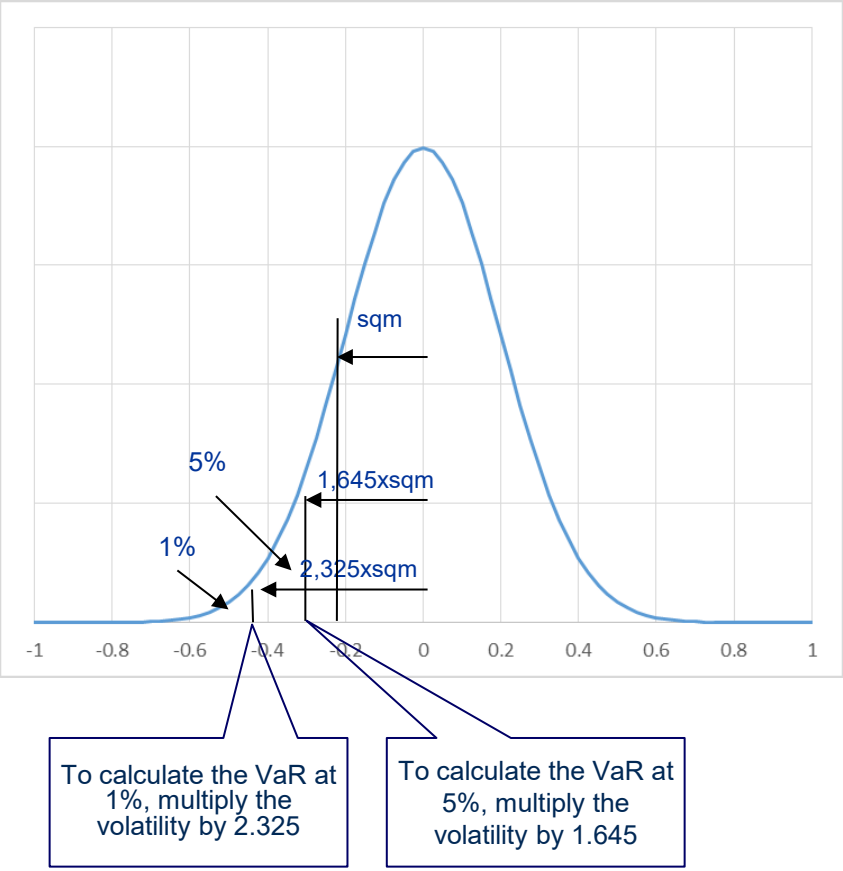


TYPICAL LEVELS

- We can associate each probability level that defines the left tail with a different multiple by which to multiply the volatility to calculate the Value at Risk

- The value defining the left tail at 5% of the distribution is equal to 1.645 times the standard deviation: 1.645×20%=32.9%, corresponding to the VaR of a unit investment at 5% probability

- Similarly, 2.325×20%=46.5%, and 1.96×20%=39.2%



VAR AND INITIAL CAPITAL

- To move on from the maximum loss on a unit investment to the maximum loss on an investment of L euro, simply multiply the VaR previously obtained by the amount of the investment

- In the example above, if the initial investment is 1,000 Euros, the Var becomes 329 Euros with probability 5% and 465 Euros with probability 1%

VAR AND TIME HORIZON

- To move from the maximum loss over a certain time horizon to the maximum loss over another time horizon, volatility must be transformed using the 'square root rule': according to this rule, the volatility on the reference horizon used for the estimation, e.g. 1 year, must be multiplied by the square root of the horizon of interest

- The rule arises from the fact that if the returns are independent in each period, the variance of a sum is the sum of the variances. For example, if the annual return is the sum of the monthly returns (which is true with the definition of logarithmic returns) we have $r_A = r_{m,1} + \cdots + r_{m,12}$ and therefore $Vr_A = V\left(r_{m,1} + \cdots + r_{m,12}\right) = 12 \times \sigma^2$ where σ^2 is the variance of monthly return. The volatility of annual return is $\sqrt{12} \times \sigma$

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VAR AND TIME HORIZON

- For example, if the volatility has been estimated at an annual level, and you want to calculate the volatility corresponding to a monthly horizon, you need to multiply the annual volatility by $\sqrt{1/12}$.

- If the volatility of the annual return is 20%, the volatility over a monthly horizon is equal to 20% x $\sqrt{1/12}$ = 5,77%. On the other hand, volatility over a 2-year horizon is $\sqrt{2} \times 20\%$=28,3%. Corresponding to intuition, volatility increases with the time horizon, albeit in a less than linear way.

QUESTIONS

- Question: What is the daily volatility of a portfolio with annual volatility of 20% assuming there are 250 days in a year?

- Question: Calculate the daily Value at Risk (with probability 2.5%) of an investment of € 15 million in a portfolio with zero expected return and annual volatility of 20%

ANSWERS

- Question: What is the daily volatility of a portfolio with annual volatility of 20% assuming there are 250 days in a year?

$0{,}2 / \sqrt{250} = 1{,}265\%$

- Question: Calculate the daily Value at Risk (with probability 2.5%) of an investment of € 15 million in a portfolio with zero expected return and annual volatility of 20%

The benchmark multiple for volatility is 1.96. The volatility to be used on a daily level is the one found in the previous question, so the VaR (in millions of euros) is 1.96 X 0.01265 X 15 = €371,883.8. In a day on a portfolio worth € 15 million you can lose a maximum of € 371,883.8 with a probability of 2.5%, i.e. you lose more than this sum only in 2.5% of cases

VAR SUMMING-UP

- Given an investment L = 1,000 Euro, with an annual return volatility of 20%, the Value at Risk over a 2-year horizon and a probability of 5% is therefore equal to $1.645 \times \sqrt{2} \times 20\% \times 1{,}000 = 465$ Euro

- Once the volatility multple corresponding to the probability you intend to leave in the left tail of the distribution has been calculated, the general rule for calculating value at Risk is therefore as follows:

multiple of volatility x square root of time horizon x volatility x amount invested

QUESTION

- Question:

(a) calculate the annual VaR (with probability 1%) of an investment of € 25 million in a portfolio with expected return 0% and annual volatility of 20%.

(b) Calculate the two-year VaR with the same parameters

(c) How do the answers change if the value of the investment is € 35 million?

Answer:

(a) The reference multiple for volatility is 2.325. The volatility is 20%, so the VaR over a one-year horizon (in millions of euros) is 2.325 X 0.2 X 25 = €11,625,000.

(b) Over a two-year horizon, multiply the VaR obtained under a) by the square root of 2 to obtain € 16,440,232.6

© multiply by 35 and divide by 25

VAR ESTIMATION

- The crucial parameter for estimating Value at Risk is volatility Statistically, estimating volatility means estimating the standard deviation of the distribution, which can be done starting from the time series of returns Financial institutions typically calculate Value at Risk over a daily horizon and directly estimate the volatility of the daily return from the time series of daily returns.

- To calculate the VaR over a weekly horizon starting from daily return data, following the rule of the square root stated above, multiply the estimate of daily volatility by the square root by 5

- In theory, the longer the time series used for the estimate, the greater the accuracy of the estimation, although structural changes must be taken into account

VOLATILITY: EMPIRICAL EVIDENCE

- Volatility has very different long-term values between asset classes

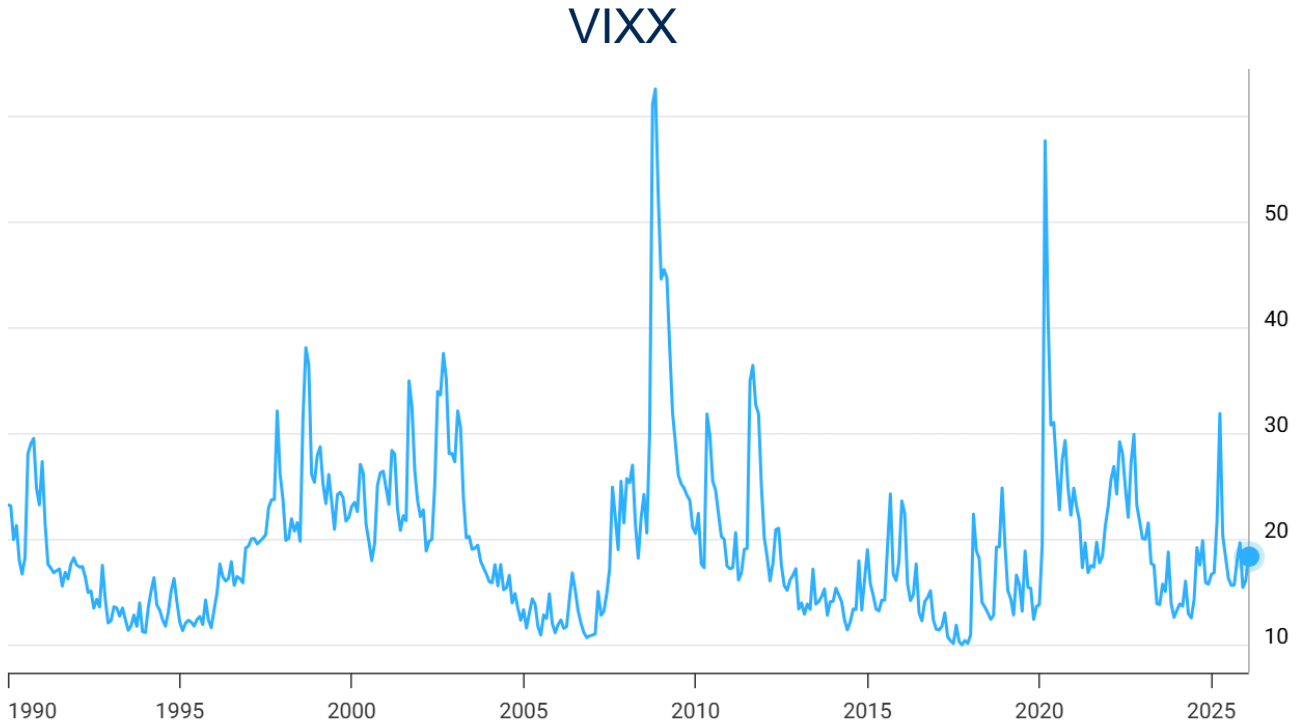
According to Dimson, Marsh and Staunton, in the last century the volatility of the real return on the stock market in the United States and England was 20%, compared to values of about 30% for Italy and Germany

- The volatility of the bond yield in real terms over the same period was

10% (US), 15% (UK), 14% (Italy), 16% (Germany)

VOLATILITY: EMPIRICAL EVIDENCE

- Volatility changes over time, as shown by the graph showing the trend of volatility estimated by the options market for the United States (VIX index) from 1990 to 2018, equal to an average of 19.3% but with peaks upwards of 70% (in October 2008 after the bankruptcy of Lehman Brothers) and downwards of up to 10%



VOLATILITY: EMPIRICAL EVIDENCE

- Volatility tends to increase during financial crisis/recession: in 1998 due to the crisis of the hedge fund Long Term Capital Management, between 2001 and 2002 during the prolonged fall in the prices of technology stocks, in 2011-2012 during the crisis in the peripheral European countries that affected markets around the world

- Over time, volatility has a tendency to return to the long-term average. The graph in the previous slide shows that the phases in which there are peaks of volatility are followed by a period of decline in volatility towards long-term values. Bob Engle won a Nobel Prize for inventing the GARCH model, which is able to explain the temporal variations of the volatility of returns in financial markets.

EWMA

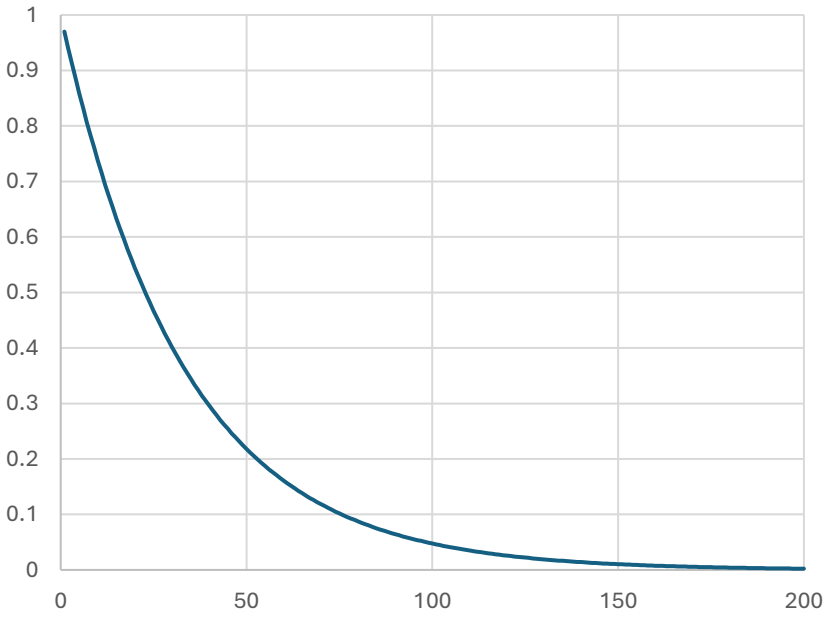
- To give greater emphasis to the most recent observations, weights are assigned that give more weight to the most recent observations and less weight to the older ones (EWMA, i.e. weights that decrease exponentially for the older observations)

$$\sigma_t^2 = \sum_{i=0}^{\infty} (1-\lambda)\lambda^i r_{t-1-i}^2$$

- λ is the decay factor. J.P.Morgan/Reuters (technical document, 1996) suggest using a decay factor of 0.94 when using daily data and 0.97 when using weekly data.

EWMA

- When 0.94 is used, it can be shown that the number of historical returns used for the estimate is 75. When 0.97 is used, the number is 150 (see the graph for the 0.97 case)



THE DYNAMICS OF VARIANCE

- As seen in the definition above, we consider the volatility estimate for day t and then assume that the volatility value changes every day

It can be shown that the previous equation can also be rewritten in the following way:

$$\sigma_t^2 = \lambda \times \sigma_{t-1}^2 + (1-\lambda) r_{t-1}^2$$

- which shows that the variance of a certain period is a weighted average of the variance of the previous period and the square of the return of the same period. The equation cannot be used for estimation of variance because the value at time t-1 is not known

THE DYNAMICS OF VARIANCE

- To obtain the EWMA model equation from this equation it is sufficient to consider the same equation for moment t-1:

$$\sigma_{t-1}^2 = \lambda \times \sigma_{t-2}^2 + (1-\lambda) r_{t-2}^2$$

- and substitute to get:

$$\sigma_t^2 = (1-\lambda) r_{t-1}^2 + \lambda(1-\lambda) r_{t-2}^2 + \lambda^2 \sigma_{t-2}^2$$

- keep substituting and find the equation in the previous page

THE DYNAMICS OF VARIANCE: COMMENTS

- Return shocks (positive or negative since in the formulas what counts is the square) significantly increase the volatility and therefore the Value at Risk of the financial portfolio

- Given the persistence implied by the model, VaR remains elevated for as long as it takes for volatility to return to its long-term value. From the equation we see that a shock of say 10% on a given day has an impact of $0.94 \times 0.1 = 0.094$ on the variance of the next day, of $0.94^2 \times 0.1 = 0.088$ on the variance of the next day, with an impact gradually decreasing

THE DYNAMICS OF VARIANCE: COMMENTS

- "no news is good news": in the absence of news, as will be seen in a later chapter, it is more likely that the return will be small in absolute value, and therefore that it will contribute to lower volatility. The absence of news, and therefore high returns in absolute value, helps to decrease volatility and therefore risk

- An alternative way to estimate volatility is to use the VIXX index, derived from the price of options: in the presence of strong volatility shocks, this method is better able to capture the dynamics of volatility, which instead takes longer with the use of the EWMA model